

BioHarvest-Sim: Analytical Assessment Report

Species: Pacific Salmon | Generated: 2026-09-07 12:02

ISO/IEC/IEEE 12207 Compliance | High-Precision Linear Algebra Core

EXECUTIVE SUMMARY:

This report details the demographic stability and sustainable yield optimization for 'Pacific Salmon' based on discrete-time Leslie matrix theory. The population exhibits a dominant eigenvalue $\lambda_1 = 1.0885$, corresponding to an intrinsic growth rate of $r = 0.0848$ per year. The net reproductive rate R_0 is 1.4040 with a mean generation time of 4.00 years. The theoretical maximum uniform sustainable harvest fraction is 8.13% per annum.

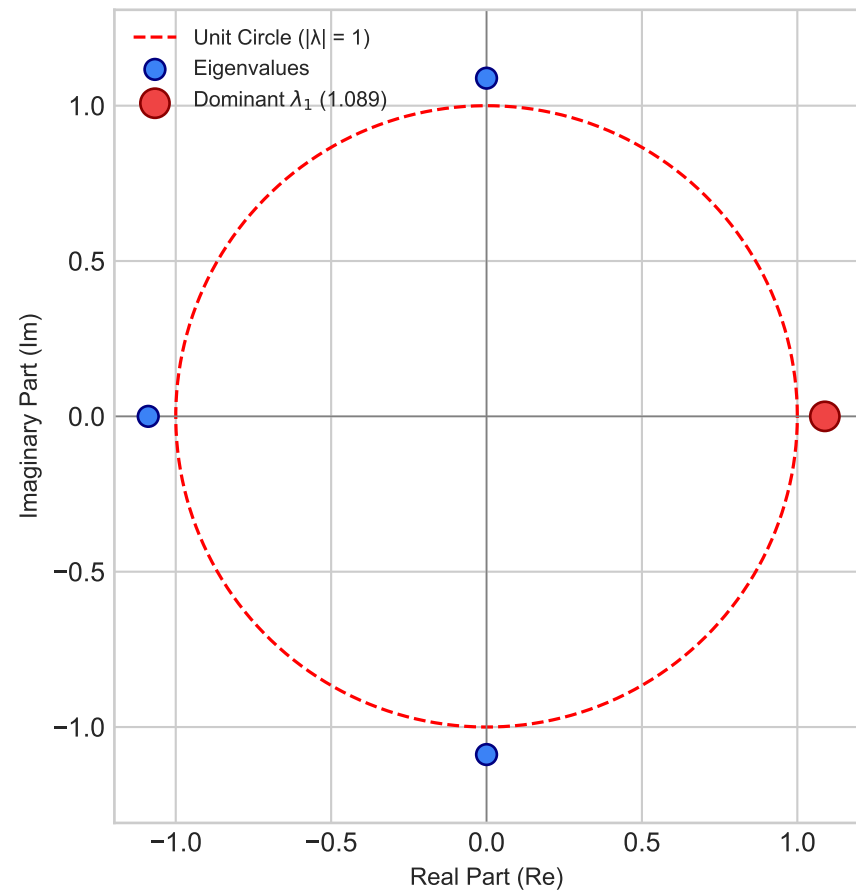
Age Classes (N)	4
Dominant Eigenvalue (λ_1)	1.08853
Intrinsic Growth Rate ($r = \ln \lambda_1$)	0.08483 yr ⁻¹
Net Reproductive Rate (R_0)	1.40400
Mean Generation Time (Tc)	4.00 years
Damping Ratio ($\lambda_1 / \lambda_2 $)	1.0000
Perron-Frobenius Primitivity	Imprimitive / Cyclic
Matrix Diagonalizable	Yes (L = P D P ⁻¹)
Uniform Sustainable Harvest (h*)	8.13% per step

Leslie Projection Matrix (L):

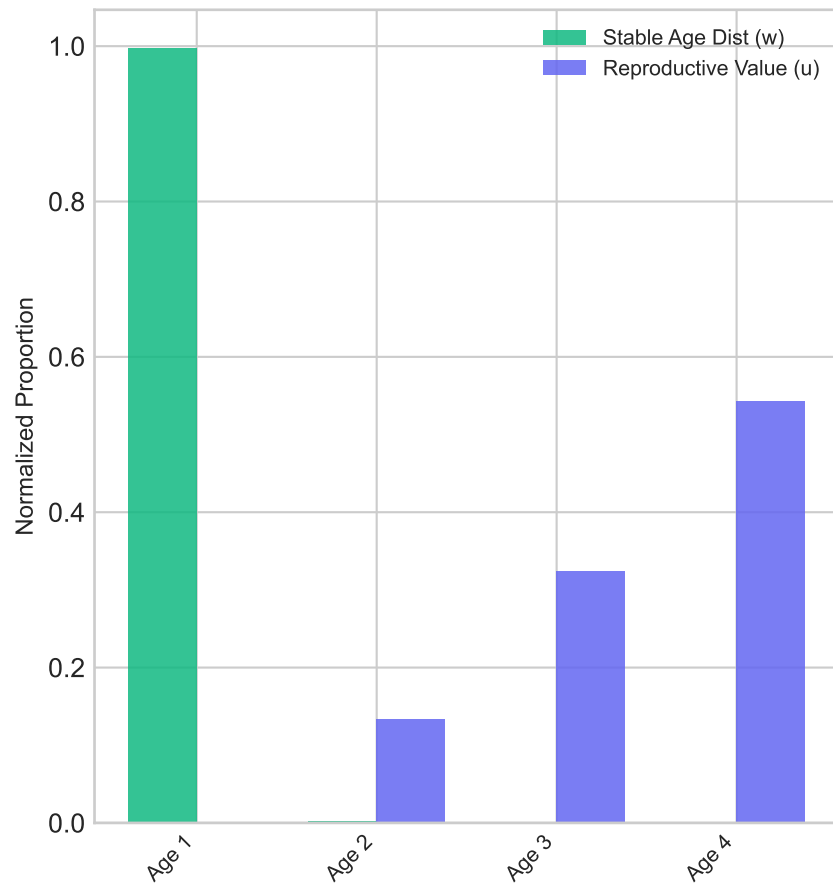
[	[	0.	0.	0.	2400.	]
[	0.002	0.	0.	0.	0.	]
[	0.	0.45	0.	0.	0.	]
[	0.	0.	0.65	0.	0.	]]

Spectral Decomposition & Asymptotic Age Structure

Eigen-Spectrum (Complex Plane)

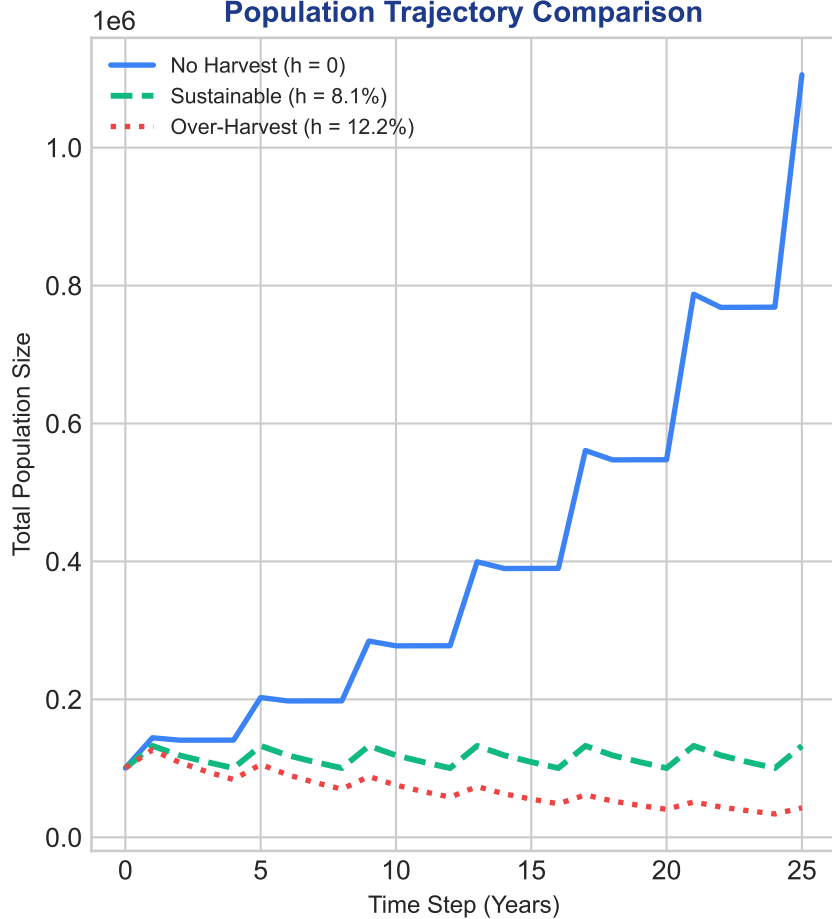


Stable Distribution vs. Reproductive Value

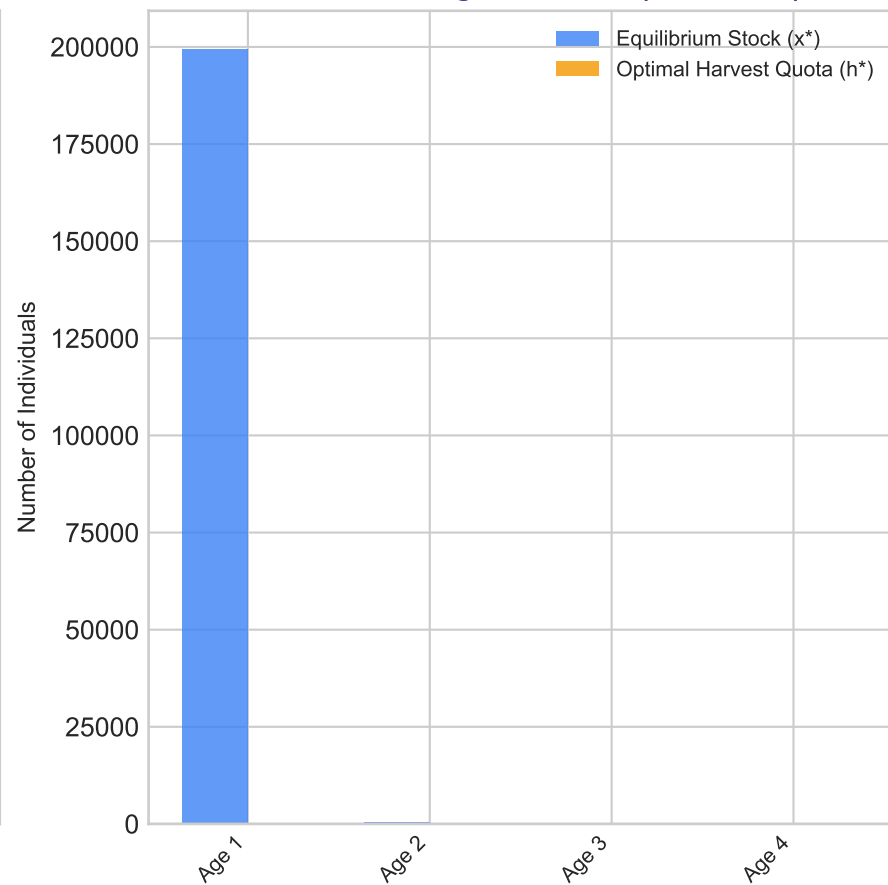


Harvest Dynamics & Optimization

Population Trajectory Comparison

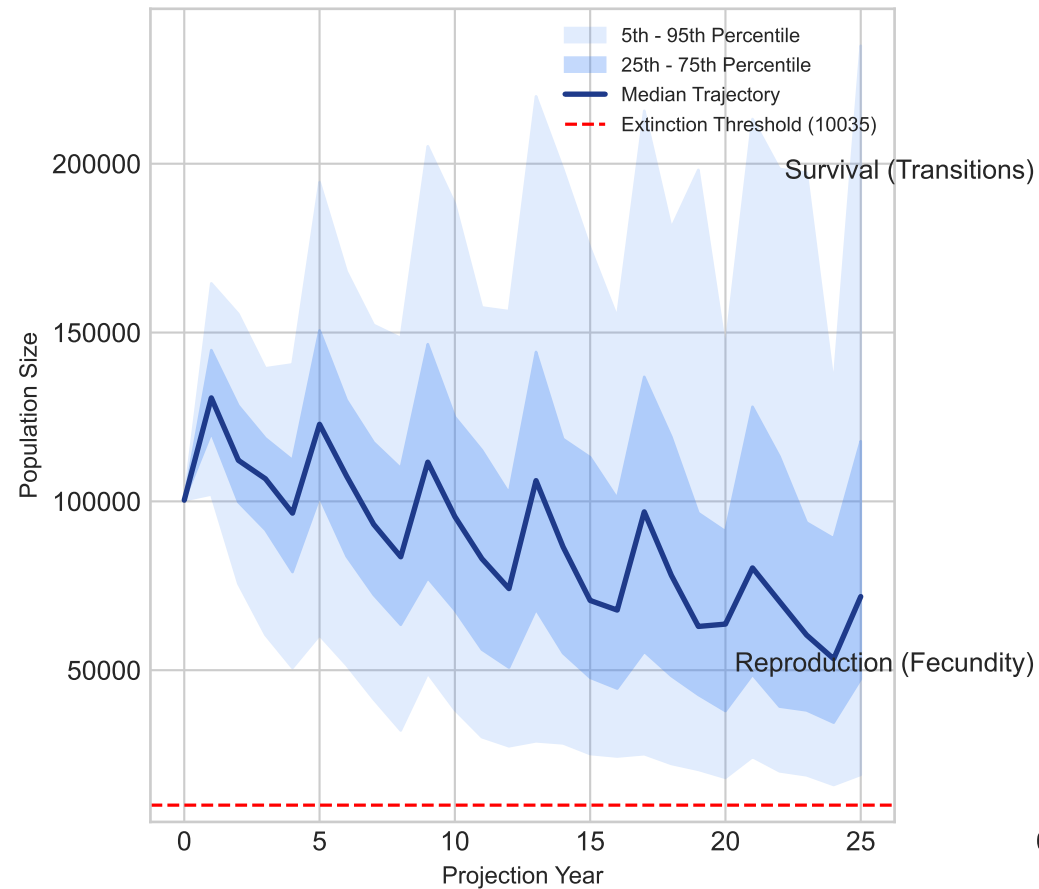


MSY Linear Program Solved (Yield: 33.6)



Environmental Stochasticity & Vital Rate Elasticities

Monte Carlo Extinction Risk (3.5%)



Elasticity Contribution to Growth (λ_1)

